

Introduction to the Lecture

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2025.SS

MCI





1. Introduction

Introduction

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Content Preview



- The goal of this lecture is ...
- This lecture is a total of 3 SWS with a total of forty-five (45) UE.
- A unit (UE) is defined as 45 min lecture.



- Lecture materials and all possible supplements will be present in its Github Repo.
 - You can easily access the link to the web-page from [here](#).

Github is chosen for easy access to material management and CI/CD capabilities and allowing hosting websites.

- In the lecture content is also distributed as a WebBook which can be accessed from the [Repo website](#).



- The student should have ...
- After completion, the student will have ...

Requirements	Taught Lecture	Code	Degree	Outcome
C/C++ Programming	Programming II	PRO2	B.Sc	ROS 2 Programming
Python Programming	Software Design	SWD	B.Sc	Linux Experience
-	-	IOT	B.Sc	Programming
-	-	-	-	-
-	-	-	-	-

Table 1: Distribution of materials across the semester.



Description	Value
Official Name	Autonome mobile Robotik
Lecture Code	AMR
Module Code	MECH-M-DUAL-3-AMR
Lecture Name	Autonomous Mobile Robotics
Semester	3
Season	WS
Lecturer	Daniel T. McGuinness, Ph.D
Module Responsible	BnM interim
Software	Linux, ROS, Python
SWS Total	3
UE Total	45
ECTS	5
Work Language	English



Assignment Type	Value
Personal Assignment A	50
Personal Assignment B	50
Sum	100



Title
A very informal journey through ROS 2
A Concise Introduction to Robot Programming with ROS2
Programming Principles and Practice using C++

Table 2: Lecture sources which can be useful during the course of the lecture. For more information on sources, please consult the [repo](#).



Topic	Units	Self Study
Locomotion	4	8
Mobile Robot Kinematics	4	8
Theory of Probability	4	8
Statistical Methods	4	8
Mobile Robot Localisation	4	8
Welcome to Linux	2	8
Command Line Fundamentals	4	8
Working with Files and Folders	2	8
Installation	1	8



Topic	Units	Self Study
ROS Concepts	4	8
Command Line Tools	4	8
Client Libraries	4	8
Transform Library	4	8
SUM	45	90